

Gate Smashers Cloud Computing Playlist Exam Mapping Guide

Course Pattern: TE IT - 2024 Pattern (SPPU)
Document Type: Exam Playlist Mapping

Academic Term: Semester V (2026)
Course Code: PEC-321A-IT

Introduction and Purpose

This document provides an exam-focused, structured mapping of the Gate Smashers Cloud Computing YouTube playlist to the official SPPU TE IT (2024 Pattern) Cloud Computing Syllabus (PEC-321A-IT). It classifies all 44 videos from the playlist into the core syllabus units.

Exam Preparation Note: Lectures highlighted in **bold** represent high-yield exam topics that are explicitly tested in SPPU end-semester exams and fully covered in the unit-wise theory notes (such as IaaS/PaaS/SaaS Service Models, Public/Private/Hybrid Deployment Models, Hypervisors, Docker & Containers, CAP Theorem, AWS/Azure Services, and RBAC Security). Focus on these lectures first for maximum marks!

Quick Statistics:

- **Total Mapped Playlist Videos:** 44 (Total Duration: 6h 46m)
- **Unit 1 (Fundamentals of Cloud Computing):** 9 Videos (1h 27m)
- **Unit 2 (Virtualization and Containerization):** 9 Videos (1h 16m)
- **Unit 3 (Cloud Platforms and Services):** 8 Videos (1h 26m)
- **Unit 4 (Cloud Security and Risk Management):** 8 Videos (1h 8m)
- **Unit 5 (Modern Cloud Environment and Emerging Technologies):** 6 Videos (52m)
- **General Overview & Career Foundations (General Overview & Career Foundations):** 4 Videos (34m)

Unit 1: Fundamentals of Cloud Computing (1h 27m)

Syllabus Topics Covered: Introduction to Cloud Computing, Need & Economic Drivers (CapEx vs OpEx), Evolution of Cloud Computing, NIST 5 Essential Characteristics, Core Advantages & Limitations, Cloud Service Models (IaaS, PaaS, SaaS), Cloud Deployment Models (Public, Private, Hybrid, Community Cloud), Cloud Architecture, Cloud Storage & Data Management.

Lecture Topic / Title	Duration	Watch Link
L-1.1: Lec-2: What is Cloud Computing Evolution and Applications Easiest introduction in Hindi	21:40	Play ↗
L-1.2: Lec-3 : Cloud Computing vs Others Real life Example	8:03	Play ↗
L-1.3: Lec-4: Introduction to Cloud Computing with Real Life Examples Key Characteristics & Benefits	8:45	Play ↗
L-1.4: Lec-8 : Cloud Architecturewith Real life examples Cloud Computing Architecture	7:49	Play ↗
L-1.5: Lec-9 : Cloud Computing Services Models - Saas, Paas and IaaS explained in Hindi	10:11	Play ↗
L-1.6: Lec-10 : SaaS, PaaS & IaaS - Cloud Computing Made Simple!	8:28	Play ↗

Lecture Topic / Title	Duration	Watch Link
L-1.7: Lec-11: Various Standards & Frameworks in Cloud Computing	7:51	Play ↗
L-1.8: Lec-13 : Storage in Cloud I Microsoft Azure Storage Services with Real life examples	9:34	Play ↗
L-1.9: Lec - 24 : Cost Models: Understanding CapEx vs. OpEx, consumption-based Pricing	5:32	Play ↗

Unit 2: Virtualization and Containerization (1h 16m)

Syllabus Topics Covered: Introduction to Virtualization, Benefits and Applications, Types of Virtualization (Full, Para, OS-level, Network, Storage, VDI), Hypervisors (Type-I Bare-Metal vs Type-II Hosted), Virtual Machines & VMware Ecosystem, VM vs Containers, Docker Fundamentals (Engine, Images, Containers), Basic Container Deployment, Introduction to Kubernetes Architecture.

Lecture Topic / Title	Duration	Watch Link
L-2.1: Lec -14 : OS Virtualization with Examples I Virtualization Types with Real life example	6:49	Play ↗
L-2.2: Lec-15 : Type 1 vs Type 2 Virtualization Explained: A Complete Guide for Cloud Computing Beginners	5:25	Play ↗
L-2.3: Lec-16 : What is Hypervisor I Types of Hypervisor	6:28	Play ↗
L-2.4: Lec-17: Introduction to VMware & Hypervisor Ecosystem with Real life Applications I Cloud Computing	8:48	Play ↗
L-2.5: Lec-18 : Virtual Machine vs Containers I Cloud Computing	9:52	Play ↗
L-2.6: Lec-35 : What is Docker Best Explanation with Real Life Examples	9:46	Play ↗
L-2.7: Lec-36 : Best Docker Practice for Beginners I How Containers run in Docker with Implementation	10:20	Play ↗
L-2.8: Lec-37 : Image vs Container with Real Life Example & Demo	6:48	Play ↗
L-2.9: Lec-38 : What is Kubernetes? Full Architecturewith Real Life Examples	12:18	Play ↗

Unit 3: Cloud Platforms and Services (1h 26m)

Syllabus Topics Covered: AWS Cloud Components, Amazon EC2 and S3 Concepts, Microsoft Azure Overview & VNet, Networking in Cloud (CIDR Addressing, Subnetting, Ports), Cloud Infrastructure Provisioning (EC2 Live Demo), Proxy Servers & Nginx Web Server Deployment.

Lecture Topic / Title	Duration	Watch Link
L-3.1: Lec-12 : Components of AWS Cloud I Cloud Computing Components	13:32	Play ↗

Lecture Topic / Title	Duration	Watch Link
L-3.2: Lec –19 : Amazon AWS VPC & Microsoft Azure VNet Cloud Network with Examples	4:50	Play ↗
L-3.3: Lec - 20 : What is CIDR Addressing Classless Inter-Domain Routing with Examples Most Important	14:16	Play ↗
L-3.4: Lec –21: Subnet in CIDR addressing with Easiest Example Subnet in IP Addressing	15:27	Play ↗
L-3.5: Lec-22 : What are Ports in Networking Why Ports are Used?	9:34	Play ↗
L-3.6: Lec-31 : How to Create Instance on AWS EC2 Amazon EC2 Explained with Live Demo	12:09	Play ↗
L-3.7: Lec-32 : What is Proxy Server & Reverse Proxy? Best Explanation with Real Life Examples	8:23	Play ↗
L-3.8: Lec-33 : What is Nginx? Best Explanation with Real Life Examples	8:33	Play ↗

Unit 4: Cloud Security and Risk Management (1h 8m)

Syllabus Topics Covered: Service Level Agreements (SLA), Data Privacy & Compliance (DPDP Act 2023), Cloud Migration (7-Step Model & Migration Challenges), OAuth 2.0 Authentication, Cloud Security Threats & Mitigation, Endpoints & API Security, Role-Based Access Control (RBAC vs ABAC).

Lecture Topic / Title	Duration	Watch Link
L-4.1: Lec-23 : SLA: Service Level Agreement in Cloud Computing Need & Use of SLA	5:18	Play ↗
L-4.2: Lec-25 : New Rules in Digital Personal Data Protection Act-2023	10:17	Play ↗
L-4.3: Lec-26 : Seven 7 Step Model of Cloud Migration Cloud Computing for Beginners	11:09	Play ↗
L-4.4: Lec-27 : Challenge in Cloud Migration Why it happens	9:06	Play ↗
L-4.5: Lec-41 : OAuth 2.0 Explained Simply! How Login with Google & Facebook Works?	6:08	Play ↗
L-4.6: Lec-42 : Cloud Security Threats and Mitigation Strategies	11:35	Play ↗
L-4.7: Lec-43 : What Are Endpoints? Master API & Cloud Security in Minutes!	6:43	Play ↗
L-4.8: Lec-44 : Introduction to Role-based access control (RBAC) RBAC vs. ABAC	8:05	Play ↗

Unit 5: Modern Cloud Environment and Emerging Technologies (52m)

Syllabus Topics Covered: Future Trends in Cloud, AI as a Service (AlaaS), Multi-Cloud & Hybrid Cloud, Serverless Computing (AWS Lambda vs EC2), Microservices Architecture vs Monolithic, Version Control (Git & GitHub), Software Development to Cloud Deployment Pipeline.

Lecture Topic / Title	Duration	Watch Link
L-5.1: Lec-5: Future of Cloud Computing AI as a Service Artificial Intelligence as a Service	5:23	Play ↗
L-5.2: Lec-28 : What is Multi Cloud with Examples Cloud Computing for Beginners	6:44	Play ↗
L-5.3: Lec-29 : What is Serverless? AWS Lambda vs EC2 Serverless Vs Server Based	9:08	Play ↗
L-5.4: Lec-30 : What are MicroServices & its Uses Microservice Architecture vs Monolithic with examples	10:34	Play ↗
L-5.5: Lec-34 : What is Version Control Git & GitHub with examples	9:08	Play ↗
L-5.6: Lec-39 : How Development Deployment is Done Pre Virtualization Era	11:26	Play ↗

General Overview & Career Foundations (34m)

Introductory, Syllabus & Career Preparation Lectures: Complete Syllabus Analysis, Cloud Roadmap for Beginners, Career Options, and AWS Certified Cloud Practitioner Guide.

Lecture Topic / Title	Duration	Watch Link
L-0.1: Lec- 1 : Complete Cloud Computing Syllabus Cloud Computing for Interviews, College	10:26	Play ↗
L-0.2: Lec-6 : Complete Roadmap Cloud Computing for Beginners How to Master in Cloud Computing	8:45	Play ↗
L-0.3: Lec-7: CareerOptions in Cloud Computing Cloud Computing for Beginners	7:09	Play ↗
L-0.4: Lec-40 : Cloud Beginner Certification Certified Cloud Practitioner Best Way to make Career!	7:50	Play ↗